

TUTORIAL NO. 3: INSTALLATION AND CONFIGURATION OF AN NMS TO MONITOR A SIMULATED NETWORK IN GNS3

Prerequisite:

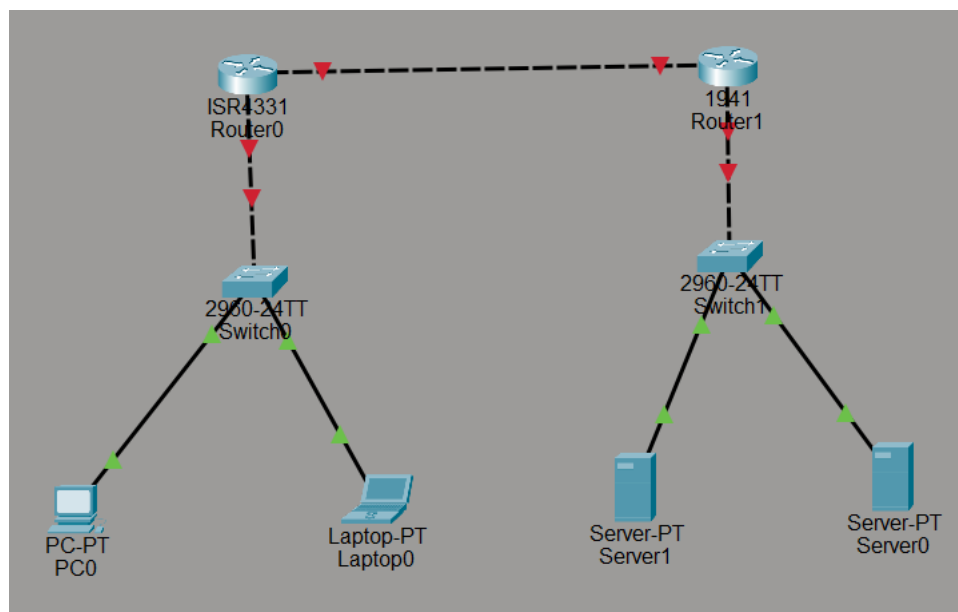
- Installation of Oracle Virtual box
- Installation of GNS3

Part 1: Installing the NMS software in Virtualization environment using Oracle VM

- 1) We need to create a virtual machine in Oracle Virtual box,
- 2) Install Windows on the VM. To install PRTG in the network, we went to its website at <https://paessler.com>, and from there download and install the software.
- 3) To allow the VM to connect to the network, a new interface needs to be added in the Network Panel of the VM. There, we disable the NAT, create a second network interface and select Virtual Box Host Only Adapter.

Part 2: Network node creation and interconnection

Consider the network architecture of an organization below:



Network Configurations

Subnet the network address 192.168.1.0/24 to create four subnetworks. One subnet will be for:

- the link between Router 1 and Router 0
- VLAN 10 containing the servers (The DMZ zone)
- VLAN 20 containing the laptop (The internal network)
- VLAN 30 containing the PC as our NMS server (Our NOC department)

Configure static route(s) between Router 0 and Router 1 (if necessary).

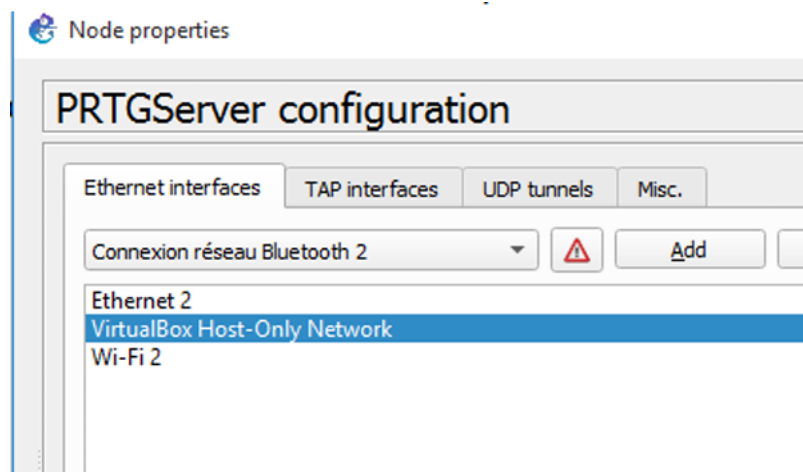
Create an Inter-VLAN between the VLANs 10, 20 and 30 to ensure communication is possible across all devices in the network.

Perform the following SNMP Configurations where necessary:

- ***snmp-server community public RO*** # to configure the public community string for read-only
- ***snmp-server community private RW*** # to configure the public community string for read-only
- ***snmp-server host <server_ip> version 2c public*** # directs the device where to send the traps (i.e. to monitoring server).
- ***snmp-server enable traps*** # enables the device to send trap messages to the monitoring server

Part 03: Connecting the PRTG Server to GNS3

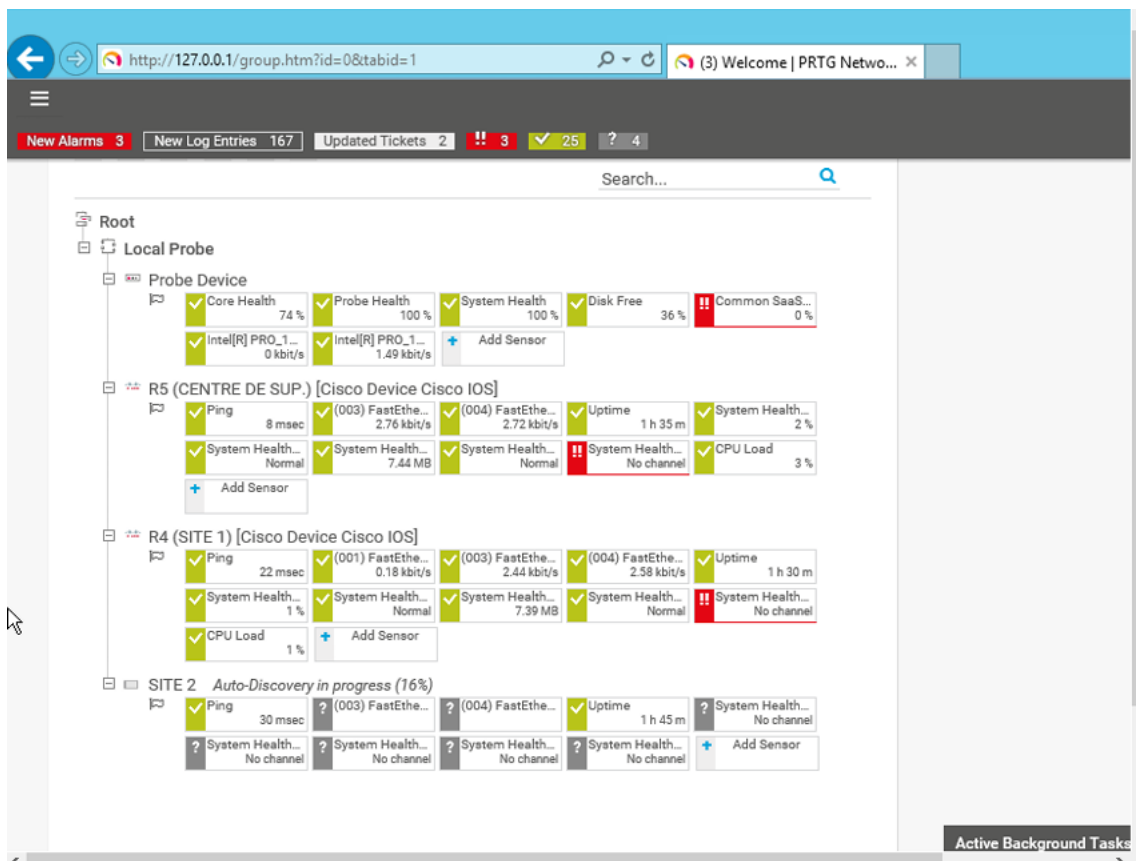
Inside our GNS3 Architecture, we connect PC0 to represent the PRTG server, and then the network interface of the virtual machine is selected and configured.



Ping different IP addresses of the network from the server, to ensure connectivity of the server to the network.

After connecting the various elements of the network as planned, and making sure all IP address are well configured to their respective ports, also making sure that the SNMP traps have been configured on all routers, in PRTG we open the PRTG Administrator page which appears in a web browser.

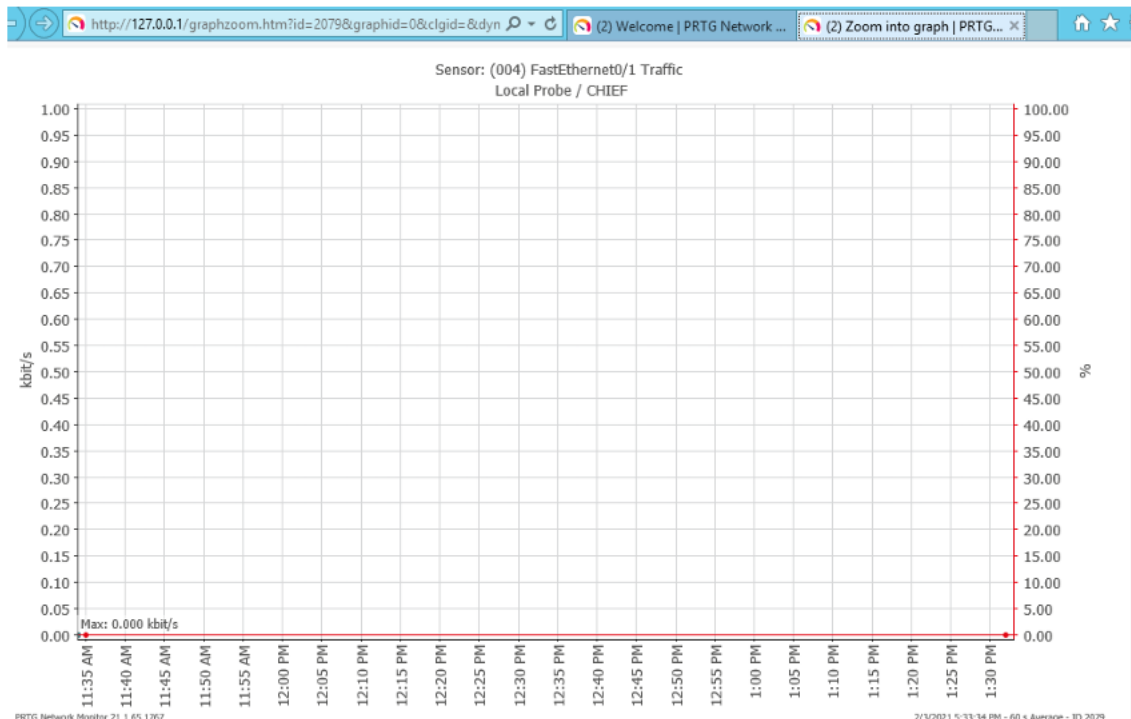
Then from there we add new devices. Each device has the IP address of a router in order to make sure the sensors can check the data packets flowing to it. When all the devices have been added, the final step is to run Auto Discovery in order to view the output from the sensors. You should obtain the results like below:



A monitored device dashboard looks like this:

Pos	Sensor	Status	Message	Graph	Priority	
1.	Ping	Up	OK	Ping Time 7 msec	★★★★★	☐
2.	(003) FastEthernet0/0 Traffic	Up	OK	Traffic Total 0.77 kbit/s	★★★★☆	☐
3.	(004) FastEthernet0/1 Traffic	Up	OK	Traffic Total 0.19 kbit/s	★★★★☆	☐
4.	Uptime	Up	OK	System Uptime 3 h 16 m	★★★★☆	☐
5.	System Health CPU	Up	OK	CPU 1 1 %	★★★★☆	☐
6.	System Health Fans	Up	OK	Fan 1 State (l) Normal	★★★★☆	☐
7.	System Health Memory	Up	OK	Available Mem 7.20 MB	★★★★☆	☐
8.	System Health Power Supplies	Up	OK	Power Supply Normal	★★★★☆	☐
9.	System Health Temperatures	Down	The sensor query was not successful be...		★★★★☆	☐
10.	CPU Load	Up	OK	CPU Load 2 %	★★★★☆	☐

Traffic analysis on a particular link can be seen as shown below:



Part III: Installing CACTI – specialized tool for network traffic monitoring.

Step 1: Download CACTI software and install on the same NMS VM having PRTG

Step 2: Use the discovery functionality of CACTI to add network nodes and links (similar to what was done with PRTG)

Step 3: Open a specific link and see the traffic statistics on that link.

Graphs

Reporting

Logs

Main Console

Create

Management

Devices

Sites

Trees

Graphs

Data Sources

Aggregates

Data Collection

Templates

Automation

Presets

Import/Export

Configuration

Utilities

Troubleshooting

Devices

Site

Any

Data Collector

Any

Template

Any

Location

All

Go

Clear

Export

Search

Enter a search term

Q

Status

Any

Devices

18

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1

2

3

4

Next

Device Description	Hostname	ID	Graphs	Data Sources	Status	In State	Uptime	Poll Time	Current (ms)	Average (ms)	Availability	Created
Cacti Server	localhost	1	4	5	Up	N/A	N/A	0.1	0	0	100 %	2020-09-06 21:43:06
Central NAS	192.168.11.105	56	12	19	Up	120	42	0.26	0.35	1.15	99.36 %	2020-09-06 21:43:06
HP Printer	192.168.11.174	55	22	22	Up	137	54	0.65	1.04	1.8	99.81 %	2020-09-06 21:43:06
vhost01	192.168.11.201	46	12	19	Up	120	4	0.38	1.45	1.61	99.99 %	2020-09-06 21:43:06
vhost02	192.168.11.202	45	12	19	Up	120	4	0.34	0.56	0.94	99.99 %	2020-09-06 21:43:06
vhost03	192.168.11.203	44	12	19	Up	120	4	0.24	0.9	2.09	99.98 %	2020-09-06 21:43:06
vhost04	192.168.11.204	43	12	19	Up	120	4	0.26	1.01	0.76	100 %	2020-09-06 21:43:06
vhost05	192.168.11.205	42	12	19	Up	120	4	0.33	0.83	1.25	99.99 %	2020-09-06 21:43:06
vhost06	192.168.11.206	41	12	19	Up	120	4	0.39	0.74	0.79	100 %	2020-09-06 21:43:06
vhost07	192.168.11.207	40	12	19	Up	267	4	0.4	0.52	1.06	98.93 %	2020-09-06 21:43:06
vhost08	192.168.11.208	39	12	19	Up	120	4	0.19	0.89	1.24	99.99 %	2020-09-06 21:43:06

